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Inventor(s)	Dayton; Ian M. et al.

Diagnostic circuit

Abstract

A diagnostic circuit includes a superconducting quantum interference device (SQUID) configured to be coupled to a test circuit such that an electronic noise present in the test circuit induces a first current that flows through the SQUID. The diagnostic circuit also includes a current path configured to receive a second current that flows through the SQUID. The SQUID is configured to generate an output in a form of: a first voltage in response to a sum of the first current and the second current being less than a threshold current and a second voltage in response to the sum of the first current and the second current being greater than the threshold current.

Inventors: Dayton; Ian M. (Arlington, VA), Nowakowski; Mark Edward (Arlington, VA)

Applicant: The Boeing Company (Arlington, VA)

Family ID: 93845213

Assignee: The Boeing Company (Arlington, VA)

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Primary Examiner: Astacio-Oquendo; Giovanni

Attorney, Agent or Firm: Harrity & Harrity, LLP

Background/Summary

FIELD

(1) The present disclosure generally relates to a diagnostic circuit, and more specifically to a diagnostic circuit configured to analyze electronic noise in a test circuit.

BACKGROUND

(2) In some applications, complementary metal-oxide-semiconductor (CMOS) circuits operate at cryogenic temperatures (e.g., less than or equal to 4 K corresponding to liquid helium or less than or equal to 77 K corresponding to liquid nitrogen). Identifying sources of electronic noise is helpful in troubleshooting such cryogenic circuits, however current methods of analyzing noise from cryogenic circuits have some disadvantages. Conventional methods involve collecting signals in the cryogenic environment and inspecting them for electronic noise with equipment that is operating at room temperature. The process of transmitting signals from the cryogenic environment to room temperature introduces more thermal noise. As a result, the noise from the cryogenic environment can be difficult to isolate from the noise from the room temperature environment or from transmission line noise along the thermal gradient between the cryogenic and room

temperature environments. Accordingly, a need exists for improved diagnostic circuits and improved diagnostic methods for analyzing electronic noise in cryogenic environments.

SUMMARY

(3) One aspect of the disclosure is a diagnostic circuit comprising: a superconducting quantum interference device (SQUID) configured to be coupled to a test circuit such that an electronic noise present in the test circuit induces a first current that flows through the SQUID; and a current path configured to receive a second current that flows through the SQUID, wherein the SQUID is configured to generate an output in a form of: a first voltage in response to a sum of the first current and the second current being less than a threshold current, and a second voltage in response to the sum of the first current and the second current being greater than the threshold current.

(4) Another aspect of the disclosure is a method of operating a diagnostic circuit, the method comprising: (a) generating a first current within a superconducting quantum interference device (SQUID) in response to the SQUID detecting an electronic noise within a test circuit; (b) increasing, while generating the first current, a magnitude of a second current that flows through the SQUID; and (c) determining the magnitude of the second current at which an output of the SQUID transitions from a first voltage to a second voltage.

(5) By the term “about” or “substantially” with reference to amounts or measurement values described herein, it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

(6) The features, functions, and advantages that have been discussed can be achieved independently in various examples or may be combined in yet other examples further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, as well as a preferred mode of use, further objectives and descriptions thereof, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying Figures.

(2) FIG. 1 is a schematic diagram of a diagnostic circuit and a test circuit, according to an example.

(3) FIG. 2 shows a Gaussian distribution of magnitudes of currents that cause a transition of a SQUID, according to an example.

(4) FIG. 3 shows a Gaussian distribution and a white noise distribution of magnitudes of currents that cause a transition of a SQUID, according to an example.

(5) FIG. 4 shows a Gaussian distribution and a shot noise distribution of magnitudes of currents that cause a transition of a SQUID, according to an example.

(6) FIG. 5 shows a Gaussian distribution and a shot noise distribution of magnitudes of currents that cause a transition of a SQUID, according to an example.

(7) FIG. 6 is a block diagram of a diagnostic device, according to an example.

(8) FIG. 7 is a schematic diagram of a diagnostic device and a test circuit, including an inset diagram of a SQUID, according to an example.

(9) FIG. 8 is a block diagram of a diagnostic apparatus, according to an example.

(10) FIG. 9 is a schematic diagram of a diagnostic system that includes a diagnostic apparatus, according to an example.

(11) FIG. 10 is a schematic diagram of a diagnostic system that includes a diagnostic apparatus,

according to an example, according to an example.

(12) FIG. **11** is a block diagram of a method, according to an example.

(13) FIG. **12** is a block diagram of a method, according to an example.

(14) FIG. **13** is a block diagram of a method, according to an example.

(15) FIG. **14** is a block diagram of a method, according to an example.

(16) FIG. **15** is a block diagram of a method, according to an example.

(17) FIG. **16** is a block diagram of a method, according to an example.

(18) FIG. **17** is a block diagram of a method, according to an example.

(19) FIG. **18** is a block diagram of a method, according to an example.

(20) FIG. **19** is a block diagram of a method, according to an example.

(21) FIG. **20** is a block diagram of a method, according to an example.

(22) FIG. **21** is a block diagram of a method, according to an example.

(23) FIG. **22** is a block diagram of a method, according to an example.

(24) FIG. **23** is a block diagram of a method, according to an example.

(25) FIG. **24** is a block diagram of a method, according to an example.

DETAILED DESCRIPTION

(26) As noted above, a need exists for improved diagnostic circuits and improved diagnostic methods for analyzing electronic noise in cryogenic environments, such as the circuits and methods included in this disclosure.

(27) A diagnostic circuit includes a superconducting quantum interference device (SQUID) configured to be (e.g., inductively) coupled to a test circuit such that an electronic noise present in the test circuit induces a first current that flows through the SQUID. The SQUID typically includes two Josephson junctions connected together to form a loop around which the first current flows. The test circuit generally takes the form of a CMOS circuit operating at a cryogenic temperature. The diagnostic circuit also includes a current path configured to receive a second current that flows through the SQUID. The SQUID is configured to generate an output in a form of: a first voltage in response to a sum of the first current and the second current being less than a threshold current, and a second voltage in response to the sum of the first current and the second current being greater than the threshold current.

(28) A method of operating the diagnostic circuit includes generating the first current within the SQUID in response to the SQUID detecting the electronic noise within the test circuit. The method also includes increasing, while generating the first current, a magnitude of the second current that flows through the SQUID. The method also includes determining the magnitude of the second current at which an output of the SQUID transitions from the first voltage to the second voltage.

(29) Accordingly, the output of the SQUID can indicate whether the sum of the first current induced by the electronic noise of the test circuit and the second current flowing through the current path is greater than or less than the threshold current of the SQUID. By controlling the magnitude of the second current flowing through the current path and noting the magnitude of the second current at which the output of the SQUID transitions, one can infer the magnitude of the first current that is induced by the electronic noise of the test circuit. By repeating this process many times, one can use statistical techniques to identify the character of the electronic noise of the test circuit. Thermal noise introduced by the room temperature analysis environment does not obscure the information indicated by the binary output of the SQUID.

(30) Disclosed examples will now be described more fully hereinafter with reference to the accompanying Drawings, in which some, but not all of the disclosed examples are shown. Indeed, several different examples may be described and should not be construed as limited to the examples set forth herein. Rather, these examples are described so that this disclosure will be thorough and complete and will fully convey the scope of the disclosure to those skilled in the art.

(31) FIG. **1** is a schematic diagram of a diagnostic circuit **10** and a test circuit **14**. The diagnostic circuit **10** includes a SQUID **12** that is (e.g., inductively) coupled to the test circuit **14** such that an

electronic noise **15** present in the test circuit **14** induces a current **16A** that flows through the SQUID **12**. The diagnostic circuit **10** also includes a current path **18** configured to receive a current **16B** that flows through the SQUID **12**. The SQUID **12** is configured to generate an output $V+$, $V-$ in a form of: a first voltage in response to a sum of the current **16A** and the current **16B** being less than a threshold current, and a second voltage in response to the sum of the current **16A** and the current **16B** being greater than the threshold current. The threshold current is typically on the order of microamps.

(32) The SQUID **12** includes a Josephson junction **11A** and a Josephson junction **11B**. The SQUID **12** forms a superconducting loop that includes two different superconducting materials that meet at the Josephson junction **11A** and the Josephson junction **11B**. The length, width, shape, and cross sectional areas of the Josephson junction **11A** and the Josephson junction **11B**, the choice of materials that form the Josephson junction **11A** and the Josephson junction **11B**, and the operating temperature of the SQUID **12** all amount to design variables or constraints that determine the threshold current that corresponds to the SQUID **12**.

(33) The current path **18** is generally made of a superconducting material that is the same as one of the materials that forms the Josephson junction **11A** and the Josephson junction **11B**.

(34) As shown, a test lead **21** is laid out in the vicinity of one more components of the test circuit **14** to capture the electronic noise **15** in the form of an induced electric current. The electronic noise **15** further induces the current **16A** that flows within the loop of the SQUID **12**. For the sake of clarity, only the current **16A** is shown circulating the loop of the SQUID **12**. However, portions of the current **16B** transit the loop as well between entering and exiting the loop. The current **16B** is generally controlled and provided by an external current source.

(35) As noted above, the SQUID **12** will generate the output $V+$, $V-$ in the form of a first voltage (e.g., 0 V) if the sum of the current **16A** and the current **16B** is less than the threshold current. The SQUID **12** will generate the output $V+$, $V-$ in the form of a second voltage (e.g., 5 V) if the sum of the current **16A** and the current **16B** is greater than the threshold current. The SQUID **12** generates the output $V+$, $V-$ between a terminal **17A** and a terminal **17B** of the SQUID **12**. Typically, the first voltage corresponding to the sum of the current **16A** and the current **16B** being less than the threshold current is less than the second voltage corresponding to the sum of the current **16A** and the current **16B** being greater than the threshold current. In various examples, the test circuit **14** (e.g., a CMOS circuit) is operated in an environment cooled by liquid nitrogen and/or liquid helium.

(36) In operation, the diagnostic circuit **10** generates the current **16A** within the SQUID **12** in response to the SQUID **12** detecting the electronic noise **15** within the test circuit **14**. While generating the current **16A**, the diagnostic circuit **10** or an external current source increases a magnitude of the current **16B** that flows through the SQUID **12**. Additionally, a computing device and/or a voltmeter determines the magnitude of the current **16B** at which the output $V+$, $V-$ of the SQUID **12** transitions from the first voltage to the second voltage.

(37) In some examples, the magnitude of the current **16B** is increased at a linear rate (or another controlled rate) with respect to time while the diagnostic circuit **10** generates the current **16A** within the SQUID **12**. A computing device and/or a voltmeter determines a time at which the output $V+$, $V-$ of the SQUID **12** transitions from the first voltage to the second voltage. The computing device then uses a magnitude of the current **16B** at the beginning of the ramp process, the known ramp rate, and the time elapsed between the start of the ramp to the transition to infer what the magnitude of the current **16B** is when the SQUID **12** transitions from the first voltage to the second voltage.

(38) In some examples, the process of increasing the current **16B** while generating the current **16A** and determining the magnitude of the current **16B** at which the output $V_{\text{sub.}+}$, $V_{\text{sub.}-}$ transitions from a first voltage to a second voltage is repeated multiple times. For example, the process could be repeated hundreds or thousands of times to obtain a statistically significant sample of data. That

is, the obtained data can be used to determine an average value, a median value, a variance, or other metrics characterizing the electronic noise **15**.

(39) In some examples, the process of increasing the current **16B** while generating the current **16A** and determining the magnitude of the current **16B** at which the output $V_{sub,+}$, $V_{sub,-}$ transitions from a first voltage to a second voltage is repeated multiple times using different ramp rates for the current **16B**. For example, each process could use a successively increasing ramp rate for the current **16B**. Varying the ramp rate of the current **16B** can be useful for detecting electronic noise **15** of the test circuit **14** that occur at different varying frequencies.

(40) FIG. 2 shows a Gaussian distribution of magnitudes of the current **16B** that cause a transition of the SQUID **12**. In this example, the electronic noise **15** primarily takes the form of thermal noise. In this example, a magnitude of about 100 μA is the most common magnitude of the current **16B** that causes a transition of the SQUID **12**. If, at a given instant, the electronic noise **15** and the current **16A** have a high value, a lesser magnitude of the current **16B** will cause the output $V_{sub,+}$, $V_{sub,-}$ to transition from the first voltage to the second voltage. On the other hand, if the electronic noise **15** and the current **16A** have a low value, a greater magnitude of the current **16B** will cause the output $V_{sub,+}$, $V_{sub,-}$ to transition from the first voltage to the second voltage.

(41) In some examples, it is useful to differentiate magnitudes of the current **16B** that correspond to a Gaussian distribution and magnitudes of the current **16B** that do not correspond to a Gaussian distribution. In FIG. 2, all of the magnitudes of the current **16B** correspond to a Gaussian distribution which indicates that thermal activity is the primary cause of the electronic noise **15**.

(42) FIG. 3 shows another distribution of magnitudes of the current **16B** that cause a transition of the SQUID **12**. In this example, there is a Gaussian component **61** and a non-Gaussian component **62**. The Gaussian component **61** represents thermal noise of the test circuit **14** whereas the non-Gaussian component **62** represents thermal noise of the diagnostic circuit **10** in the form of white noise.

(43) FIG. 4 shows another distribution of magnitudes of the current **16B** that cause a transition of the SQUID **12**. In this example, there is a Gaussian component **61** and a non-Gaussian component **62**. The Gaussian component **61** represents thermal noise of the test circuit **14** whereas the non-Gaussian component **62** represents thermal noise of the diagnostic circuit **10** in the form of a single source of shot noise.

(44) FIG. 5 shows another distribution of magnitudes of the current **16B** that cause a transition of the SQUID **12**. In this example, there is a Gaussian component **61** and a non-Gaussian component **62**. The Gaussian component **61** represents thermal noise of the test circuit **14** whereas the non-Gaussian component **62** represents thermal noise of the diagnostic circuit **10** in the form of multiple sources of shot noise.

(45) FIG. 6 is a block diagram of a diagnostic device **20**. The diagnostic device **20** includes a probe **22**, a motor **24**, a SQUID **12**, and a current path **18**.

(46) FIG. 7 is a schematic diagram of the diagnostic device **20** and the test circuit **14**, including an inset diagram of the SQUID **12**.

(47) The motor **24** is configured to scan the probe **22** over the test circuit **14**. The diagnostic device **20** also includes the SQUID **12** that is attached to the probe **22** and is configured to be (e.g., inductively) coupled to the test circuit **14** as the motor **24** moves the probe **22** to different positions over the test circuit **14**.

(48) Depending on the location of the probe **22** over the test circuit **14**, the electronic noise **15A** or the electronic noise **15B** present in the test circuit **14** induces the current **16A** that flows through the SQUID **12**. The diagnostic device **20** also includes the current path **18** configured to receive the current **16B** that flows through the SQUID **12**. The SQUID **12** is configured to generate the output $V+$, $V-$ in the form of: the first voltage in response to the sum of the current **16A** and the current **16B** being less than the threshold current, and a second voltage in response to the sum of the current **16A** and the current **16B** being greater than the threshold current.

(49) The probe **22** generally takes the form of any probe used for scanning probe microscopy. The motor **24** generally takes the form of any motor that can control the position of the probe **22** in three dimensions with respect to the test circuit **14**. The probe **22** can be scanned over the test circuit **14** using the motor **24** to analyze electronic noise **15** at different positions of the test circuit **14**. That is, the motor **24** is configured to move the probe **22** along the x-axis and the y-axis that are orthogonal to each other and parallel to the test circuit **14**. The motor **24** is also configured to move the probe **22** along the z-axis that is orthogonal to the x-axis, the y-axis, and the test circuit **14**.

(50) As shown, the SQUID **12** is scanned along with the probe **22** by the motor **24** to capture the electronic noise **15** in the form of an induced electric current. The electronic noise **15** further induces the current **16A** that flows within the loop of the SQUID **12**. The current **16B** is generally controlled and provided by an external current source that is operatively coupled to the diagnostic device **20**.

(51) In operation, the motor **24** moves the probe **22** over a location **19A** of the test circuit **14** such that the SQUID **12** generates the current **16A** within the SQUID **12** in response to the SQUID **12** detecting the electronic noise **15A** at the location **19A** of the test circuit **14**. The current **16B** is increased via an external current source while the current **16A** is generated. A computing device determines the magnitude of the current **16B** at which the output $V+$, $V-$ of the SQUID **12** transitions from the first voltage to the second voltage while the probe **22** was over the location **19A**.

(52) Next, the motor **24** moves the probe **22** over the location **19B** of the test circuit **14** such that the SQUID **12** generates the current **16A** in response to the SQUID **12** detecting the electronic noise **15B** at the location **19B** of the test circuit **14**. The current **16B** is increased via the external current source while the current **16A** is generated. The computing device determines the magnitude of the current **16B** at which the output $V+$, $V-$ of the SQUID **12** transitions from the first voltage to the second voltage while the probe **22** was over the location **19B**.

(53) As noted above with reference to FIGS. 2-5, the process of increasing the current **16B** while generating the current **16A** and determining the magnitude of the current **16B** at which the output $V_{sub. +}$, $V_{sub. -}$ transitions from a first voltage to a second voltage can be repeated multiple times for the location **19A** and multiple times for the location **19B**. For example, the process could be repeated hundreds or thousands of times for each location to obtain a statistically significant sample of data for each location. That is, the obtained data can be used to determine an average value, a median value, a variance, or other metrics characterizing the electronic noise **15** at each location. As noted above with reference to FIGS. 2-5, the magnitudes of the current **16B** that cause the transition at each location can be classified as corresponding or not corresponding to a Gaussian distribution. Likewise, the aforementioned process of using different ramp rates of the current **16B** can be used at each location as well.

(54) FIG. 8 is a block diagram of a diagnostic apparatus **30**. The diagnostic apparatus **30** includes an array **32A** of SQUIDs **12**. The diagnostic apparatus **30** also includes a current source **34**. The current source **34** can include any circuit that is controllable to generate a desired current magnitude and provide that current to another circuit.

(55) FIG. 9 is a schematic diagram of a diagnostic system **40** that includes the diagnostic apparatus **30**. The diagnostic system **40** includes an array **32B** of test circuits **14A-F** and an array **32A** of SQUIDs **12A-F**. When the array **32A** is positioned suitably over the array **32B**, each SQUID **12A-F** of the array **32A** is (e.g., inductively) coupled respectively to a test circuit **14A-F** of the array **32B** such that an electronic noise present in the test circuits **14A-F** induces the currents **16A** that flow through the respective SQUIDs **12A-F**. The diagnostic system **40** also includes the current source **34** (e.g., a multi-output current source) configured to provide the currents **16B** for each SQUID **12A-F** of the array **32A** such that the currents **16B** flow respectively through the SQUIDs **12A-F**. Each SQUID **12A-F** of the array **32A** is configured to generate its own output $V+$, $V-$ in the form of: a first voltage (e.g., 0 V) in response to a sum of the current **16A** and the current **16B** being less

than the threshold current, and a second voltage (e.g., 5 V) in response to the sum of the current **16A** and the current **16B** being greater than the threshold current.

(56) In operation, the diagnostic apparatus **30** generates, for each SQUID **12A-F** of the array **32A** and a corresponding test circuit **14A-F** of the array **32B**, the current **16A** within the SQUID **12A-F** in response to detecting an electronic noise **15** respectively within the test circuit **14A-F**. While generating the currents **16A** within the SQUIDS **12A-F**, the diagnostic apparatus **30** (e.g., the current source **34**) increases the magnitudes of the currents **16B** flowing through the respective SQUIDS **12A-F**. Additionally, a computing device and/or a voltmeter determines, for each SQUID **12A-F** of the array **32A**, the magnitude of the current **16B** at which the output $V+$, $V-$ of the SQUID **12A-F** transitions from the first voltage to the second voltage.

(57) Referring to FIG. **10**, the diagnostic apparatus **30** is used to analyze a new array of test circuits **14**. As such, the diagnostic apparatus **30** is moved to be in close proximity to an array **32C** of test circuits **14G-L**. Accordingly, the diagnostic apparatus **30** generates, for each SQUID **12A-F** of the array **32A** and a corresponding test circuit **14G-L** of the array **32C**, the current **16A** within the SQUID **12A-F** in response to detecting an electronic noise **15** respectively within the test circuits **14G-L**. While generating the currents **16A** within the SQUIDS **12A-F**, the diagnostic apparatus **30** (e.g., the current source **34**) increases the magnitudes of the currents **16B** flowing through the respective SQUIDS **12A-F**. Additionally, a computing device and/or a voltmeter determines, for each SQUID **12A-F** of the array **32A**, the magnitude of the current **16B** at which the output $V+$, $V-$ of the SQUID **12A-F** transitions from the first voltage to the second voltage.

(58) As noted above with reference to FIGS. **2-5**, the process of increasing the current **16B** while generating the current **16A** and determining the magnitude of the current **16B** at which the output V_{sub+} , V_{sub-} transitions from the first voltage to the second voltage can be repeated multiple times for each test circuit **14A-F** of the array **32B** and multiple times for each test circuit **14G-L** of the array **32C**. For example, the process could be repeated hundreds or thousands of times for each test device of each array to obtain a statistically significant sample of data for each test device of each array. That is, the obtained data can be used to determine an average value, a median value, a variance, or other metrics characterizing the electronic noise **15** at each test device of each array.

(59) As noted above with reference to FIGS. **2-5**, the respective magnitudes of the current **16B** that causes the transition of each SQUID of each array can be classified as corresponding or not corresponding to a Gaussian distribution. Likewise, the aforementioned process of using different ramp rates of the current **16B** can be used for each SQUID of each array as well.

(60) FIGS. **11-14** are block diagrams of a method **100**, a method **110**, a method **114**, and a method **118** of operating the diagnostic circuit **10**. FIGS. **15-19** are block diagrams of a method **200**, a method **218**, a method **224**, and a method **228** of operating the diagnostic device **20**. FIGS. **20-24** are block diagrams of a method **300**, a method **310**, a method **322**, a method **324**, and a method **326** for operating the diagnostic apparatus **30**. As shown in FIGS. **11-24**, the methods **200-326** include one or more operations, functions, or actions as illustrated by blocks **102**, **104**, **106**, **108**, **112**, **116**, **202**, **204**, **206**, **208**, **210**, **212**, **214**, **216**, **220**, **222**, **226**, **302**, **304**, **306**, **308**, **312**, **314**, **316**, **318**, **320**, and **328**. Although the blocks are illustrated in a sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon the desired implementation.

(61) At block **102**, the method **100** includes (a) generating the current **16A** within the SQUID **12** in response to the SQUID **12** detecting the electronic noise **15** within a test circuit **14**. Functionality related to block **102** is discussed above with reference to FIG. **1**.

(62) At block **104**, the method **100** includes (b) increasing, while generating the current **16A**, the magnitude of the current **16B** that flows through the SQUID **12**. Functionality related to block **104** is discussed above with reference to FIG. **1**.

(63) At block **106**, the method **100** includes (c) determining the magnitude of the current **16B** at

which the output $V+$, $V-$ of the SQUID 12 transitions from the first voltage to the second voltage. Functionality related to block 106 is discussed above with reference to FIGS. 1-5.

(64) At block 108, the method 110 includes identifying the magnitudes of the currents 16B that correspond to a Gaussian distribution. Functionality related to block 108 is discussed above with reference to FIGS. 2-5.

(65) At block 112, the method 114 includes identifying the magnitudes of the currents 16B that do not correspond to the Gaussian distribution. Functionality related to block 112 is discussed above with reference to FIGS. 3-5.

(66) At block 116, the method 118 includes performing steps (a), (b), and (c) a second time, wherein the second time performing step (b) comprises increasing the magnitude of the current 16B at a second rate that is greater than the first rate. Functionality related to block 116 is discussed above with reference to FIG. 1.

(67) At block 202, the method 200 includes (a) moving, via the motor 24, the probe 22 over the location 19A of the test circuit 14, where the SQUID 12 is attached to the probe 22. Functionality related to block 202 is discussed above with reference to FIG. 7.

(68) At block 204, the method 200 includes (b) generating the current 16A within the SQUID 12 in response to the SQUID 12 detecting the electronic noise 15A at the location 19A of the test circuit 14. Functionality related to block 204 is discussed above with reference to FIG. 7.

(69) At block 206, the method 200 includes (c) increasing, while generating the current 16A, the magnitude of the current 16B that flows through the SQUID 12. Functionality related to block 206 is discussed above with reference to FIG. 7.

(70) At block 208, the method 200 includes (d) determining the magnitude of the current 16B at which the output $V_{\text{sub.}+}$, $V_{\text{sub.}-}$ of the SQUID 12 transitioned from the first voltage to the second voltage while the probe 22 was over the location 19A. Functionality related to block 208 is discussed above with reference to FIG. 7.

(71) At block 210, the method 200 includes (e) moving, via the motor 24, the probe 22 over the location 19B of the test circuit 14. Functionality related to block 210 is discussed above with reference to FIG. 7.

(72) At block 212, the method 200 includes (f) generating the current 16A within the SQUID 12 in response to the SQUID 12 detecting the electronic noise 15B at the location 19B of the test circuit 14. Functionality related to block 212 is discussed above with reference to FIG. 7.

(73) At block 214, the method 200 includes (g) increasing, while generating the current 16A, the magnitude of the current 16B that flows through the SQUID 12. Functionality related to block 214 is discussed above with reference to FIG. 7.

(74) At block 216, the method 200 includes (h) determining the magnitude of the current 16B at which the output $V_{\text{sub.}+}$, $V_{\text{sub.}-}$ of the SQUID 12 transitioned from the first voltage to the second voltage while the probe 22 was over the location 19B. Functionality related to block 216 is discussed above with reference to FIG. 7.

(75) At block 220, the method 218 includes identifying the magnitudes of the currents 16B that correspond to a Gaussian distribution. Functionality related to block 220 is discussed above with reference to FIGS. 2-5.

(76) At block 222, the method 224 includes identifying the magnitudes of the currents 16B that do not correspond to the Gaussian distribution. Functionality related to block 222 is discussed above with reference to FIGS. 3-5.

(77) At block 226, the method 228 includes performing steps (b), (c), and (d) a second time, wherein the second time performing step (c) comprises increasing the magnitude of the current 16B at a second rate that is greater than the first rate. Functionality related to block 222 is discussed above with reference to FIG. 7.

(78) At block 302, the method 300 includes (a) generating, for each SQUID 12A-F of the array 32A of SQUIDS 12A-F and a corresponding test circuit 14A-F of the array 32B of test circuits 14A-F,

the current **16A** within the SQUID **12** in response to detecting the electronic noise **15** within the test circuit **14**. Functionality related to block **302** is discussed above with reference to FIG. **9**.

(79) At block **304**, the method **300** includes (b) increasing, for each SQUID **12A-F** of the array **32A**, the magnitude of the current **16B** that flows through the SQUID **12A-F** while generating the current **16A**. Functionality related to block **304** is discussed above with reference to FIG. **9**.

(80) At block **306**, the method **300** includes (c) determining, for each SQUID **12A-F** of the array **32A**, the magnitude of the current **16B** at which the output $V+$, $V-$ of the SQUID **12A-F** transitions from the first voltage to the second voltage. Functionality related to block **306** is discussed above with reference to FIG. **9**.

(81) At block **308**, the method **310** includes moving the diagnostic apparatus **30** after performing steps (a), (b), and (c). Functionality related to block **308** is discussed above with reference to FIG. **9** and FIG. **10**.

(82) At block **312**, the method **310** includes generating, for each SQUID **12A-F** of the array **32A** and a corresponding test circuit **14G-J** of the array **32C** of test circuits **14G-J**, the current **16A** within the SQUID **12A-F** in response to detecting the electronic noise **15** within the test circuit **14G-J**. Functionality related to block **312** is discussed above with reference to FIG. **10**.

(83) At block **314**, the method **310** includes increasing, for each SQUID **12A-F** of the array **32A**, the magnitude of the current **16B** that flows through the SQUID **12A-F** while generating the current **16A**. Functionality related to block **314** is discussed above with reference to FIG. **10**.

(84) At block **316**, the method **310** includes determining, for each SQUID **12A-F** of the array **32A**, the magnitude of the current **16B** at which the output $V_{sub.+}$, $V_{sub.-}$ of the SQUID **12A-F** transitions from the first voltage to the second voltage. Functionality related to block **316** is discussed above with reference to FIG. **10**.

(85) At block **318**, the method **322** includes identifying, for each test circuit **14-A-F** of the array **32B**, the magnitudes of the currents **16B** that correspond to a Gaussian distribution. Functionality related to block **318** is discussed above with reference to FIGS. **2-5**.

(86) At block **320**, the method **324** includes identifying, for each test circuit **14-A-F** of the array **32B**, the magnitudes of the currents **16B** that do not correspond to the Gaussian distribution. Functionality related to block **320** is discussed above with reference to FIGS. **3-5**.

(87) At block **328**, the method **326** includes performing steps (a), (b), and (c) a second time, wherein the second time performing step (b) comprises increasing the magnitude of the current **16B** at a second rate that is greater than the first rate. Functionality related to block **328** is discussed above with reference to FIG. **9**.

(88) The description of the different advantageous arrangements has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the examples in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different advantageous examples may describe different advantages as compared to other advantageous examples. The example or examples selected are chosen and described in order to explain the principles of the examples, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various examples with various modifications as are suited to the particular use contemplated.

Claims

1. A diagnostic circuit comprising: a superconducting quantum interference device (SQUID) configured to be coupled to a test circuit such that an electronic noise present in the test circuit induces a first current that flows through the SQUID; and a current path configured to receive a second current that flows through the SQUID, wherein the SQUID is configured to generate an output in a form of: a first voltage in response to a sum of the first current and the second current being less than a threshold current, and a second voltage in response to the sum of the first current

- and the second current being greater than the threshold current.
2. The diagnostic circuit of claim 1, wherein the test circuit is configured to operate in an environment cooled by liquid nitrogen.
 3. The diagnostic circuit of claim 1, wherein the test circuit is configured to operate in an environment cooled by liquid helium.
 4. The diagnostic circuit of claim 1, wherein the SQUID is configured to be inductively coupled to the test circuit.
 5. The diagnostic circuit of claim 1, wherein the SQUID comprises a first terminal and a second terminal and the SQUID is configured to generate the output between the first terminal and the second terminal.
 6. The diagnostic circuit of claim 1, wherein the first voltage is substantially equal to zero.
 7. The diagnostic circuit of claim 1, wherein the first voltage is less than the second voltage.
 8. The diagnostic circuit of claim 1, wherein the test circuit comprises a complementary metal-oxide semiconductor (CMOS) circuit.
 9. A method of operating a diagnostic circuit, the method comprising: (a) generating a first current within a superconducting quantum interference device (SQUID) in response to the SQUID detecting an electronic noise within a test circuit, wherein the SQUID is configured to be coupled to the test circuit such that the electronic noise present in the test circuit induces the first current that flows through the SQUID; (b) increasing, while generating the first current, a magnitude of a second current that flows through the SQUID, wherein a current path is configured to receive the second current that flows through the SQUID; and (c) determining the magnitude of the second current at which an output of the SQUID transitions from a first voltage to a second voltage, wherein the first voltage is output in response to a sum of the first current and the second current being less than a threshold current, and the second voltage is output in response to the sum of the first current and the second current being greater than the threshold current.
 10. The method of claim 9, wherein the method comprises performing steps (a), (b), and (c) at least two times.
 11. The method of claim 10, further comprising identifying the magnitudes of the second currents that correspond to a Gaussian distribution.
 12. The method of claim 11, further comprising identifying the magnitudes of the second currents that do not correspond to the Gaussian distribution.
 13. The method of claim 9, wherein increasing the magnitude of the second current comprises increasing the magnitude at a first rate, the method further comprising performing steps (a), (b), and (c) a second time, wherein the second time performing step (b) comprises increasing the magnitude at a second rate that is greater than the first rate.
 14. The method of claim 9, wherein the method is performed while the test circuit is being cooled by liquid nitrogen.
 15. The method of claim 9, wherein the method is performed while the test circuit is being cooled by liquid helium.
 16. The method of claim 9, wherein generating the first current comprises generating the first current via inductive coupling between the SQUID and the test circuit.
 17. The method of claim 9, wherein the first voltage is substantially equal to zero.
 18. The method of claim 9, wherein the first voltage is less than the second voltage.
 19. The method of claim 9, wherein the second voltage is greater than zero.
 20. The method of claim 9, wherein the test circuit comprises a complementary metal-oxide semiconductor (CMOS) circuit.
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